SUPERSTORE SALES AND PROFIT ANALYSIS

Why High Sales Do Not Always Mean High Profit

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1. Executive Summary

This report analyses a retail superstore dataset of 9,977 sales transactions to show which categories, regions and products make money, and which ones only look good on sales. The store recorded **$2,296,196** in sales and **$286,241** in profit, an overall profit margin of **12.5%**.

The main finding is that high sales do not always mean high profit. The key results are:

- **Furniture sells a lot but earns little.** It is second on sales ($741,306) but has a margin of only 2.5%, against 17.4% for Technology.
- **Tables lose money despite strong sales.** They rank 4th on sales ($206,966) but lose $17,725. Bookcases and Supplies also lose money.
- **The Central region underperforms.** It makes 21.8% of sales but only 13.9% of profit.
- **Discounts do not raise sales but reduce profit.** The correlation between discount and sales is -0.03, and between discount and profit is -0.22.

2. Data and Method

The analysis uses the SampleSuperstore dataset, with one row per transaction and 13 columns: ship mode, segment, country, city, state, postal code, region, category, sub-category, sales, quantity, discount and profit. It covers 3 categories, 17 sub-categories and 4 regions across the United States.

Cleaning: no missing values were found. 17 duplicate rows were removed, reducing 9,994 rows to 9,977 transactions. All figures in this report use the 9,977 de-duplicated transactions. Totals were grouped by category, sub-category, region, segment and ship mode, and the relationships between sales, profit, discount and quantity were measured with correlations.

3. Overall Performance

Key performance indicator	Result
Total Sales	$2,296,195.59
Total Profit	$286,241.42
Profit Margin	12.5%
Transactions	9,977
Average Sale per Transaction	$230.15
Average Profit per Transaction	$28.69

4. Findings

Finding 1: Furniture sells a lot but earns little

Technology leads on both sales ($836,154) and profit ($145,455). Furniture ranks second on sales but earns only 6.4% of total profit, with a margin of 2.5%. Office Supplies sells the least of the three but earns 42.7% of profit.

Category	Sales	Profit	Margin	Share of profit
Technology	$836,154	$145,455	17.4%	50.8%
Furniture	$741,306	$18,422	2.5%	6.4%
Office Supplies	$718,735	$122,365	17.0%	42.7%

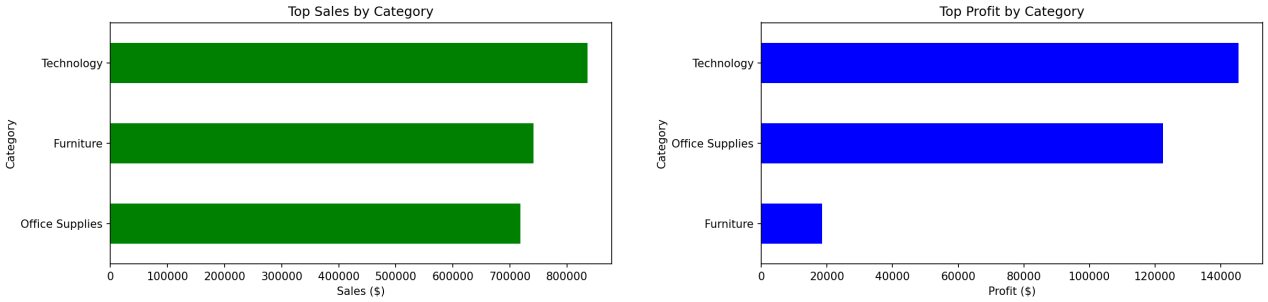


Figure 1: Sales by category (left) and profit by category (right).

Recommendation: Review Furniture supplier costs and pricing. At a 2.5% margin, a small rise in costs or discounts removes its profit entirely.

Finding 2: Central underperforms the other regions

West is the strongest region on both measures, with 31.6% of sales and 37.8% of profit. Central is the weakest: it makes 21.8% of sales but only 13.9% of profit, and its margin of 7.9% is about half of West's (14.9%). This points to cost, discount or product-mix problems in Central.

Region	Share of sales	Share of profit	Margin
West	31.6%	37.8%	14.9%
East	29.5%	32.0%	13.5%
Central	21.8%	13.9%	7.9%
South	17.1%	16.3%	11.9%

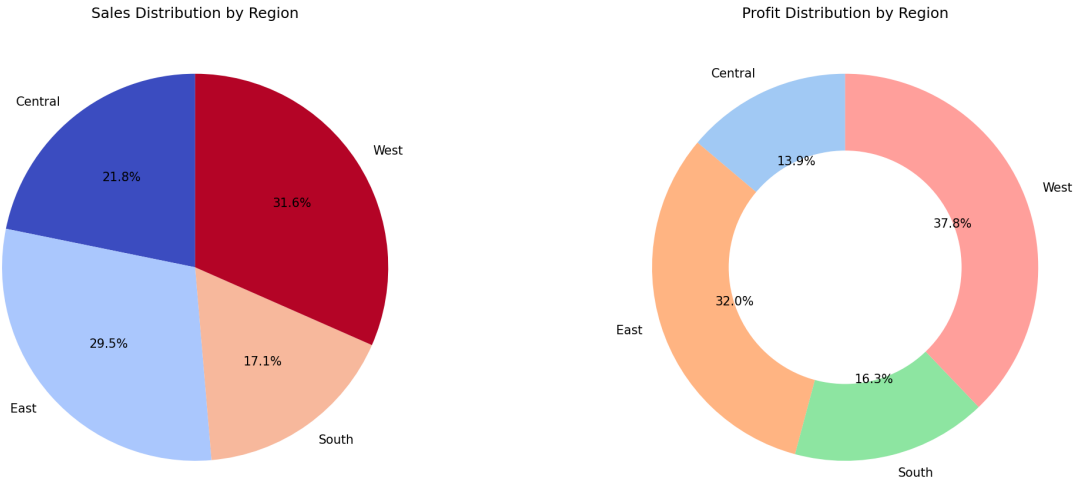


Figure 2: Share of sales by region (left) and share of profit by region (right).

Recommendation: Investigate Central's costs, discounts and product mix to find out why its margin is half of West's.

Finding 3: Tables sell well but lose money

Phones ($330,007) and Chairs ($327,778) lead on sales. Copiers earn the most profit ($55,618, a 37.2% margin), followed by Phones ($44,516) and Accessories ($41,937). Three sub-categories lose money: Tables (-$17,725), Bookcases (-$3,473) and Supplies (-$1,189). Tables are the biggest warning sign, because they rank 4th on sales yet are the least profitable product. Chairs also sell almost as much as Phones but earn a margin of only 8.1%.

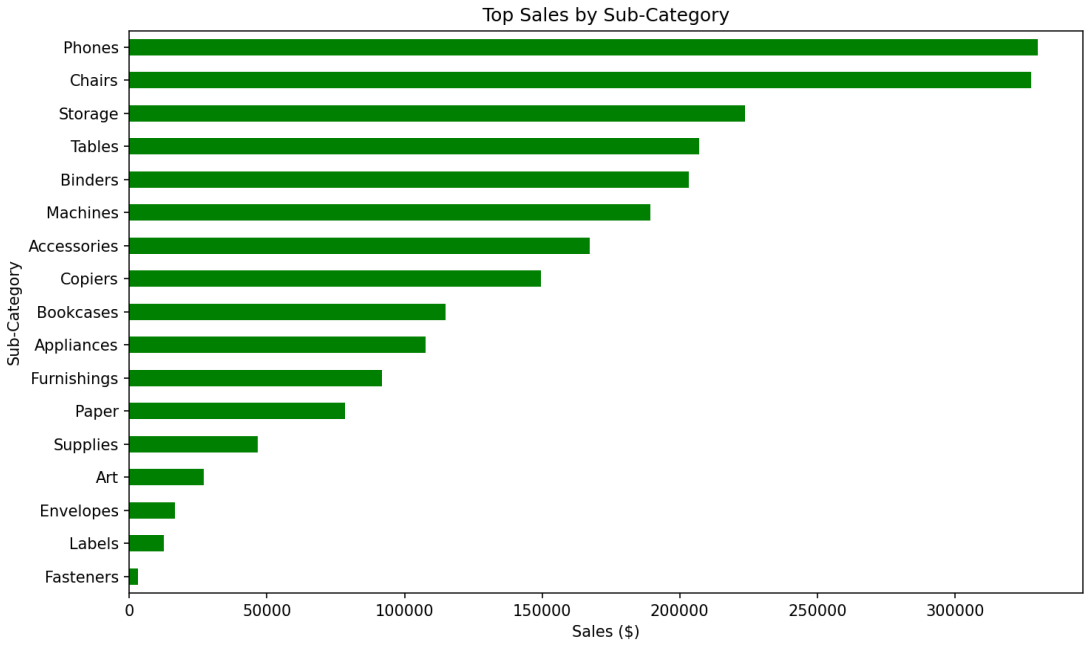


Figure 3: Sales by sub-category.

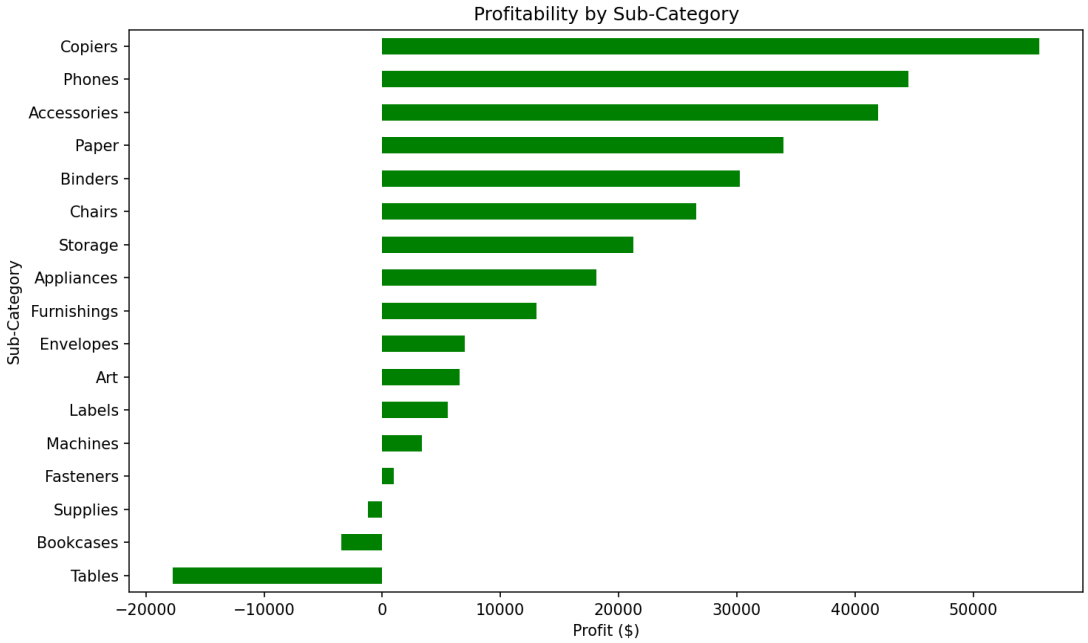


Figure 4: Profit by sub-category. Tables, Bookcases and Supplies are below zero.

Recommendation: Review Tables, Bookcases and Supplies. Check their costs and discounts, and consider repricing, cutting discounts or dropping the weakest lines. Promote Copiers, Accessories and Phones, which earn strong profits.

Finding 4: Discounts do not lift sales but reduce profit

Sales and profit are moderately linked (0.48), so higher sales generally lead to higher profit, but far from always. Sales and discount show almost no link (-0.03), so discounts do not boost sales. Profit and discount are negatively linked (-0.22), so higher discounts tend to reduce profit. Sales and quantity are only weakly linked (0.20), so revenue depends more on order value than on the number of units.

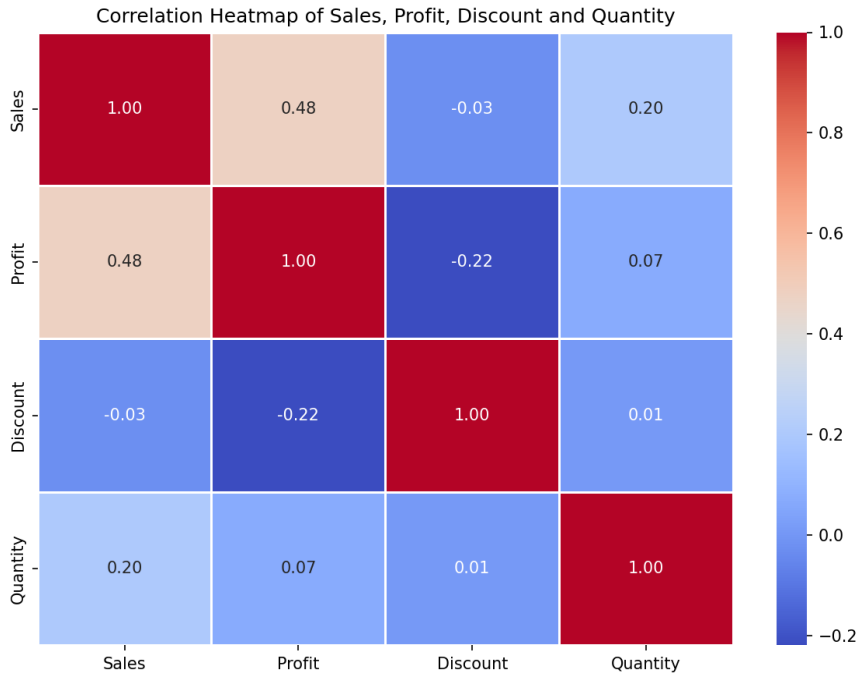


Figure 5: Correlation heatmap of sales, profit, discount and quantity.

Recommendation: Target discounts instead of applying them widely. They do not increase sales, but they are linked to lower profit.

Finding 5: Segments and ship modes show no major problems

Consumers make up 50.6% of sales but have the lowest margin of the three segments (11.5%), against 13.0% for Corporate and 14.0% for Home Office. Standard Class carries 59.1% of sales and 57.3% of profit, and the four ship modes have similar margins (12.1% to 13.9%), so none stands out.

5. Conclusion

The store is profitable overall, with a 12.5% margin on $2,296,196 of sales, but profit is not spread evenly. Technology and Office Supplies earn margins of about 17%, while Furniture earns 2.5%. West leads among the regions, while Central earns a 7.9% margin on 21.8% of sales. Tables, Bookcases and Supplies lose money, and discounts do not lift sales but are linked to lower profit. The central message is that **high sales do not always mean high profit**, so decisions should look at profit and margin as well as sales.

6. Summary of Recommendations

Area	Recommended action
Products	Review Tables, Bookcases and Supplies; reprice, cut discounts or drop the weakest lines.
Furniture	Check supplier costs and pricing, since the margin is only 2.5%.
Central region	Investigate costs, discounts and product mix to close the gap with West.
Discounts	Target them instead of applying them widely.
Growth	Promote high-margin lines such as Copiers, Accessories and Phones.

7. Assumptions and Limitations

- The dataset is a sample retail dataset used for practice, not a real company's records.
- The data has no dates, so trends over time cannot be analysed.
- Profit is taken as given in the dataset. How it was calculated is not documented.
- Correlations show relationships, not causes. A link between discounts and lower profit does not prove one causes the other.
- Category, region and product effects overlap, so a difference cannot be put down to a single factor.

Appendix: Supporting Tables

A. By Category

Category	Sales	Profit	Margin	Share of sales	Transactions
Technology	$836,154	$145,455	17.4%	36.4%	1,847
Furniture	$741,306	$18,422	2.5%	32.3%	2,118
Office Supplies	$718,735	$122,365	17.0%	31.3%	6,012

B. By Region

Region	Sales	Profit	Margin	Share of sales	Transactions
West	$725,256	$108,330	14.9%	31.6%	3,193
East	$678,435	$91,506	13.5%	29.5%	2,845
Central	$500,783	$39,656	7.9%	21.8%	2,319
South	$391,722	$46,749	11.9%	17.1%	1,620

C. By Segment

Segment	Sales	Profit	Margin	Share of sales	Transactions
Consumer	$1,160,833	$134,007	11.5%	50.6%	5,183
Corporate	$706,070	$91,955	13.0%	30.7%	3,015
Home Office	$429,293	$60,279	14.0%	18.7%	1,779

D. By Ship Mode

Ship Mode	Sales	Profit	Margin	Share of sales	Transactions
Standard Class	$1,357,316	$163,969	12.1%	59.1%	5,955
Second Class	$459,177	$57,447	12.5%	20.0%	1,943
First Class	$351,380	$48,954	13.9%	15.3%	1,537
Same Day	$128,322	$15,872	12.4%	5.6%	542

E. By Sub-Category

Sub-Category	Sales	Profit	Margin	Share of sales	Transactions
Phones	$330,007	$44,516	13.5%	14.4%	889
Chairs	$327,778	$26,567	8.1%	14.3%	615
Storage	$223,844	$21,279	9.5%	9.7%	846
Tables	$206,966	-$17,725	-8.6%	9.0%	319
Binders	$203,409	$30,228	14.9%	8.9%	1,522
Machines	$189,239	$3,385	1.8%	8.2%	115
Accessories	$167,380	$41,937	25.1%	7.3%	775
Copiers	$149,528	$55,618	37.2%	6.5%	68
Bookcases	$114,880	-$3,473	-3.0%	5.0%	228
Appliances	$107,532	$18,138	16.9%	4.7%	466
Furnishings	$91,683	$13,053	14.2%	4.0%	956
Paper	$78,224	$33,944	43.4%	3.4%	1,359
Supplies	$46,674	-$1,189	-2.5%	2.0%	190
Art	$27,107	$6,525	24.1%	1.2%	795
Envelopes	$16,476	$6,964	42.3%	0.7%	254
Labels	$12,445	$5,526	44.4%	0.5%	363
Fasteners	$3,024	$950	31.4%	0.1%	217